

TEA-CART: How-to Guide

Guidance on using the [tool](#)

As you enter data, please keep the following in mind:

- This application does not require a login and is free to use.
- In the left side panel, there is a button to Download User Inputs. It is essential that you download data from the server after long periods of inactivity. If you need to leave the application before completing the analysis, you can later upload the saved file to continue your work.
- Data are entered in tables. To begin entering (or editing) data, double click anywhere in the table using your mouse. When you are done entering data, press **CTRL + ENTER** on your keyboard (this initiates the calculation). Data can only be entered for one table at a time.
- When you are ready to see the results of your analysis, proceed to the Scenarios tab and select scenarios for analysis. Either choose groups of projects (to support a comparison) or click the button to select all projects for each scenario.

Steps to use the tool

1. Use the navigation panel at the top to select **Inputs**.



2. Within **Inputs**, select **Baseline** to choose your state and enter the years used for the planning forecast. You may also change some parameters for the forecast, the scope of emissions to include (for example, whether to include certain upstream emissions), and other assumptions.



3. Within **Projects**, enter information about each project. To begin entering (or editing) data, double click in the table using your mouse. Alternatively, if you would like to provide budget-level inputs, skip to step 5 below.

Projects 1 | Bicycle and Pedestrian Lane Miles of New Infrastructure ^

Reset Projects 1

This category represents implementation of any **two-way miles of new bicycle or pedestrian facility**. The default assumption for these project types is that any new bicycle or pedestrian facility would be two-way. For one-way facilities, please enter half the total miles for the facility.

Search:

Area Type	Facility Type	Unit	2025	2030	2050
Core	New shared-use path	New lane-miles	0	0	0
Urban	New shared-use path	New lane-miles	0	0	0
Suburban	New shared-use path	New lane-miles	0	0	0
Rural	New shared-use path	New lane-miles	0	0	0
Core	New bicycle lane	New lane-miles	0	0	0

Double-click the value (here a '0') to begin editing

4. When you are done entering data, press **CTRL + ENTER** on your keyboard (this initiates the calculation). Data can only be entered for one table at a time.

Projects 1 | Bicycle and Pedestrian Lane Miles of New Infrastructure

Reset Projects 1

This category represents implementation of any **two-way miles of new bicycle or pedestrian facility**. The default assumption for these project types is that any new bicycle or pedestrian facility would be two-way. For one-way facilities, please enter half the total miles for the facility.

Search:

Area Type	Facility Type	Unit	2025	2030	2050
Core	New shared-use path	New lane-miles	0	0	0
Urban	New shared-use path	New lane-miles	0	0	0
Suburban	New shared-use path	New lane-miles	10	20	0
Rural	New shared-use path	New lane-miles	0	0	0
Core	New bicycle lane	New lane-miles	0	0	0
Urban	New bicycle lane	New lane-miles	0	0	0

Hit CTRL + Enter to finish entering, or Esc to cancel

5. In the **Budget** tab, data can be entered the same way as in the **Projects** tab. If you already entered data in the **Projects** tab, go to step 7 below. If instead you would like to provide budget-level inputs, double click with your mouse in the Budget tables to begin entering values. Once you are done, press **CTRL + ENTER** on your keyboard to initiate the calculation. As a reminder, data can only be entered one table at a time.

Budget 1 | Bicycle and Pedestrian Lane Miles of New Infrastructure ⓘ
⌵

This category represents spending on any **two-way miles of new bicycle or pedestrian facility**. The default assumption for these project types is that any new bicycle or pedestrian facility would be two-way. For one-way facilities, please enter half the budget amount.

Reset Budget 1

Search:

Category	Area Type	Facility Type	Unit	Value
Bicycle and Pedestrian	Core	New shared-use path	New lane-miles	0
Bicycle and Pedestrian	Urban	New shared-use path	New lane-miles	0
Bicycle and Pedestrian	Suburban	New shared-use path	New lane-miles	.25
Bicycle and Pedestrian	Rural	New shared-use path	New lane-miles	0
Bicycle and Pedestrian	Core	New bicycle lane	New lane-miles	0
Bicycle and Pedestrian	Urban	New bicycle lane	New lane-miles	0

Similarly, in Budget Mode, double click cells to begin editing.

6. When entering **Budget** values, you must also update the **Projects** before beginning the calculation. Or, fill **Budget** values by updating from Projects.

Baseline

Projects

Budget

Costs

Assumptions

Scenarios

Advanced

Please provide **budget-level inputs** for one or more of the project categories shown below, keeping in mind the following two rules when entering project-level inputs:

- i) All spending is assumed to be equally distributed over the years of spending by the corresponding horizon year.
- ii) Spending estimates are assumed to be averages based on historical data.

Fill Project Tab with Budget Inputs

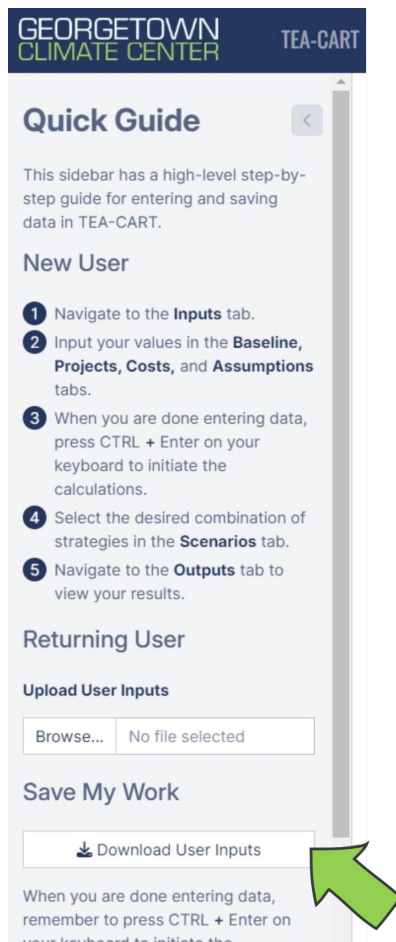
Fill Budget Tab with Project Inputs

Funding Start Year: ⓘ
2026

Total Years Covered: ⓘ
5

Total Budget (\$M) for 2026-2030: ⓘ
200

7. Regularly download the data that you have entered. You can download it by clicking the **Download User Inputs** button on the sidebar.



8. If desired, within **Costs**, enter custom unit costs for the project type. Note that default values have already been provided.

Baseline Projects Budget **Costs** Assumptions Scenarios Advanced

This section provides information on the **cost inputs** for the project categories shown below. Please click on the different fields to overwrite the default values with any custom values provided by the user.

Costs 1 | Bicycle & Pedestrian Costs Reset Costs 1

This category represents the **overall cost per mile** of bicycle or pedestrian facilities being implemented.

Search:

Area Type	Facility Types	Cost per Facility Mile	Annual Maintenance Cost per Facility Mile
Core	New shared-use path	566955	56695.5
Urban	New shared-use path	566955	56695.5
Suburban	New shared-use path	566955	56695.5
Rural	New shared-use path	566955	56695.5
Core	New bicycle lane	36258.75	3625.875
Urban	New bicycle lane	36258.75	3625.875

Default values can be changed.

9. Further **Assumptions** for the analysis are for advanced users. Data can be changed similar to the previous tabs. For more information about changing assumptions, please refer to the user guide.

Baseline Projects Budget Costs **Assumptions** Scenarios Advanced

This section provides information on the **input assumptions** for the categories shown below. These inputs affect the GHG impact and effectiveness of each strategy category. Please click on the different fields to overwrite the default values with any custom values provided by the user.

More information on the categories can be found by clicking the pull-down arrow next to each category.

Assumptions 1 | Bicycle and Pedestrian Parameters ^

Reset Assumptions 1

This category includes the following data types:

- (i) **Prior drive mode share of new walkers / bikers**, which represents the fraction of new walkers / bikers who would have previously driven a car. This is calculated as a percentage of the new walkers / bikers or "persons per square mile" (ppsm) within the different area types defined.
- (ii) **Average trip length**, which represents the average Walk and / or Bike trip length in miles.
- (iii) **Annualization**, which represents the annual number of days multiplier for the mode-shift to walking / biking.

Search:

Element	Unit	Value
Annualization	Annualization factor	365
Walk	Average trip length	0.7
Bike	Average trip length	2.3

Default values can be changed.

10. In the **Scenarios** tab, click the boxes to choose which groups of projects to include in the scenario analysis.

Baseline Projects Budget Costs Assumptions **Scenarios** Advanced

This tab provides the different possible strategies provided in the TEA-CART analysis. Please select the desired combination of strategies to be used as forecast "scenarios." The resulting output can be used to compare up to two different scenarios.

Scenario Selections

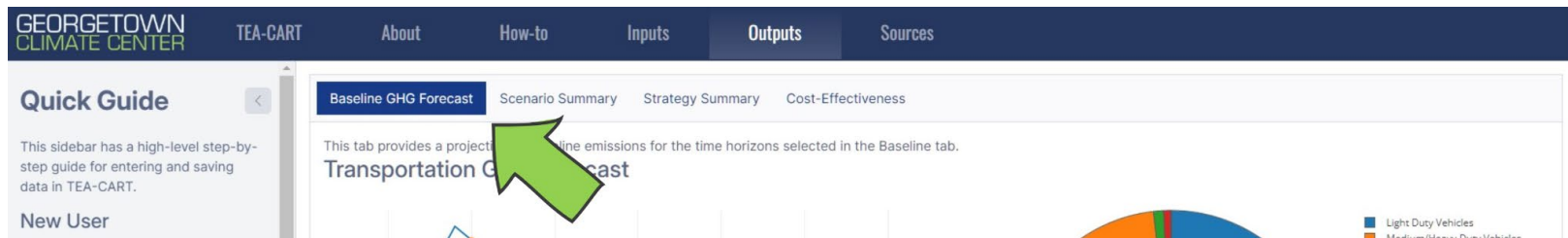
Search:

	Grouped Projects	Relation to Project Tab	Scenario1	Scenario2
1	Bicycle and Pedestrian	Projects 1	☒	☐
2	Transit Service Expansion	Projects 2	☒	☐
3	Micromobility	Projects 3	☒	☒
4	Travel Demand Management	Projects 4	☒	☒
5	Park-and-Ride	Projects 5	☒	☐

By default, all project types will be selected under Scenario 1. Click checkboxes to add or remove project types.

11. It is possible to use a custom forecast on the **Advanced** tab for the number of EVs on the road (for example, for states that have a goal to add one million EVs to the road by a certain deadline), future VMT, and other advanced parameters. Refer to the user guide for more information.

12. To see the results of the data that have been entered, refer to the **Outputs** tab.



13. Click Download Summary Report to get a PDF document with all of this information.

GEORGETOWN
CLIMATE CENTER

TEA-CART

Step guide for entering and saving data in TEA-CART.

New User

- 1 Navigate to the **Inputs** tab.
- 2 Input your values in the **Baseline, Projects, Costs, and Assumptions** tabs.
- 3 When you are done entering data, press CTRL + Enter on your keyboard to initiate the calculations.
- 4 Select the desired combination of strategies in the **Scenarios** tab.
- 5 Navigate to the **Outputs** tab to view your results.

Returning User

Upload User Inputs

Browse...

No file selected

Save My Work

Download User Inputs

When you are done entering data, remember to press CTRL + Enter on your keyboard to initiate the calculations.

Download Summary Report

More information can be found in the **How-to** tab.